

Image Fusion with Diffusion Models

1 Introduction

In recent years, diffusion-based generative models have shown their powerful ability to generate images conditioned on different texts or images [1, 2]. However, most of the diffusion models generate images conditioned on single text or single images and some diffusion models can generate images conditioned on multiple inputs while preserving the structure of one input and preserving the semantics of other inputs [2, 3].

My idea is to use diffusion models which conditioned on two or multiple images to generate images that combine the semantic characteristics of input images. This is a prompt-free approach, giving more flexibility and creativity to the model.

2 Related Work

Clip is a strong image and text encoder that maximizes the similarity of the embedding space of a pair of an image and the corresponding text. SDG finetunes a clip classifier and uses the guidance function (1) to generate images [4]. s_i are scaling factor and $F_{\phi_i(x_i, y_i, t)}$ calculate the difference between the noise image x_t and the input image y_i in clip embedding space. The original work does not use two images as input but uses one as text and another as image. However, I found out that using two images as input works as well.

$$F_{\phi_0}(x_t, y, t) = s_1 F_{\phi_1}(x_t, y_1, t) + s_2 F_{\phi_2}(x_t, y_2, t) \quad (1)$$

Interpolation is another way to combine the semantic characteristics of input images. However, it can now just deal with two input images and can not generate excellent images when the two input images are from two domains that have slight similarities [5].

Overall, not much work has studied the field of the fusion of two or multiple images. This can be seen as a bold attempt and breakthrough.

3 Method

The reason the interpolation may not work well when the two input images are completely from two different

domains is probably because the interpolation of two latent space embeddings does not guarantee to preserve the semantic information of both input images and sometimes will generate weird results. Because of this, I think it is still reasonable to use the clip to guide the diffusion model to generate the images. However, like SDG method, training an explicit clip classifier is extremely expensive [2]. So, based on the idea of Control-Net, I will possibly build an implicit clip controller which combines the embedding of the two input images by changing and finetuning the Control-Net using the following loss function[2, 6]:

$$\mathcal{L}_1 = \mathbb{E}_{z_0, t, c_t, c_f, \epsilon \sim \mathcal{N}(0,1)} [\|\epsilon - \epsilon_\theta(z_t, t, c_t, c_f)\|_2^2] \quad (2)$$

$$\mathcal{L}_2 = F_{\phi_1}(f(x_t, t), y_1, t) + F_{\phi_2}(f(x_t, t), y_2, t) \quad (3)$$

The c_t is the text condition, which can be set to *None* while training and sampling, and c_f is the combinational semantic feature of two input images. The function f is defined as follows [6]:

$$f(x_t, t) := \frac{x_t - \sqrt{1 - \alpha_t} \epsilon_\theta(x_t, t)}{\sqrt{\alpha_t}} \quad (4)$$

4 Plan of Project

4.1 Challenges

This is a relatively difficult project and study for me. A large amount of time is needed to understand the code, write my own code, and train my model. Time is also needed to better design the loss function and the clip control-net. I can anticipate that there will be even more difficulties when doing the project, but I will try my best to overcome them

4.2 Milestones

Finish reading and writing the code before Nov.19 and starting training my model at least on Nov.20. It is likely that the first time is not successful. So I will probably compare my most recent model can other's models beginning at Nov. 30. My model should generate images that combine the semantic feature of two input images

References

- [1] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models, 2021.
- [2] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models, 2023.
- [3] Senmao Li, Joost van de Weijer, Taihang Hu, Fahad Shahbaz Khan, Qibin Hou, Yaxing Wang, and Jian Yang. Stylediffusion: Prompt-embedding inversion for text-based editing. *arXiv preprint arXiv:2303.15649*, 2023.
- [4] Xihui Liu, Dong Huk Park, Samaneh Azadi, Gong Zhang, Arman Chopikyan, Yuxiao Hu, Humphrey Shi, Anna Rohrbach, and Trevor Darrell. More control for free! image synthesis with semantic diffusion guidance. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2023.
- [5] Clinton J. Wang and Polina Golland. Interpolating between images with diffusion models, 2023.
- [6] Gwanghyun Kim, Taesung Kwon, and Jong Chul Ye. Diffusionclip: Text-guided diffusion models for robust image manipulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2426–2435, June 2022.